

Where Does the Algorithmic Map Fall? BISECT vs. GerryChain Ensembles in Three States

Real Traces, Diagnostics, and Cross-Tool Disagreement

G.1 — GerryChain Congressional Comparison

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Abstract

We compare the tract-level NRS v0.1 BISECT benchmark with four-chain, 2,000-step Rust ReCom and GerryChain ensembles in Rhode Island, Iowa, and North Carolina. All inputs, traces, RPLAN/RCTX baselines, diagnostics, and hashes are archived. Rhode Island’s benchmark cut fraction falls at the 1.35th percentile in Rust and 1.81st in GerryChain. Iowa falls at 0.33rd and 0.88th percentiles, but effective sample sizes are too small for a stable sub-1% tail claim. North Carolina falls at 0–0.2%, yet pooled ESS is only 87–109 and the two implementations produce materially different cut distributions. Wisconsin was removed before sampling because its benchmark contained a disconnected district. These results replace the earlier unsupported six-state percentile tables and demonstrate why trace custody, diagnostics, and cross-tool comparison are prerequisites for any extremum claim.

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1. Real Evidence Package

The active package is `docs/examples/g-ensemble-evidence-packages/G.1-G.3+real-2020/`. It binds the BISECT baselines, 2016 and 2020 election inputs, Rust and GerryChain traces, RPLAN/RCTX contexts, audit certificates, analysis, and verification commands.

1.1. Design

The preregistered design used four independent chains and 2,000 steps per chain at 0.5% population tolerance, discarding the first 500 steps. Partition snapshots were retained every ten steps. Rust used a random-weight Kruskal tree kernel matching GerryChain 0.3.2 for this comparison [DeFord et al., 2021, MGGG Redistricting Lab, 2021]. The default Wilson UST kernel remains a distinct sampler and is not conflated with this cross-tool experiment.

Wisconsin failed the input eligibility gate: district 1 was disconnected under the sampled adjacency graph. The failure remains archived. Iowa was added as a medium-state replacement before any production chain metric was inspected.

1.2. Cut-Fraction Results

Table 1: Benchmark percentile in the unweighted tract-graph cut-fraction distribution.

State	Tool	Percentile	95% CI	$\hat{R}$	ESS	Accept.
RI	Rust	1.35	[0.93, 1.77]	1.0005	4290	0.790
RI	GerryChain	1.81	[1.48, 2.07]	1.0000	6000	1.000
IA	Rust	0.33	[0.20, 0.47]	1.0046	257	0.997
IA	GerryChain	0.88	[0.46, 1.18]	1.0005	498	0.997
NC	Rust	0.00	[0.00, 0.00]	1.0404	109	1.000
NC	GerryChain	0.20	[0.00, 0.40]	1.0070	87	0.993

Rhode Island supports a bounded low-tail result in both tools. Iowa suggests a low-tail benchmark, but neither ESS reaches the preregistered 1,000 threshold for a sub-1% headline. North Carolina cannot support a precise tail claim: GerryChain ESS is below 100 and Rust is only slightly above it.

1.3. Cross-Tool Differences

The cut distributions are not interchangeable. Two-sample Kolmogorov–Smirnov statistics are 0.038 for Rhode Island, 0.162 for Iowa, and 0.471 for North Carolina. With thousands of autocorrelated draws, the associated nominal p -values are not independent-sample legal tests; the effect sizes nevertheless show material kernel/implementation sensitivity in Iowa and especially North Carolina.

Partition-space integrated autocorrelation times from ten-step snapshots are approximately 21 for Rhode Island, 11 for Iowa, and 7–8 for North Carolina. Scalar convergence alone therefore does not imply independent partitions.

2. Claim Boundary

Cut fraction is a topology proxy, not Polsby–Popper or Reock. These three tract-level states do not establish a national compactness extremum, neutral distribution, legal validity, or block-level NRS conformance. The former Wisconsin, Georgia, Pennsylvania, Texas, California, and North Carolina percentile claims are withdrawn unless regenerated by an active package.

3. Conclusion

The strongest result is methodological. A benchmark percentile is not one number: it depends on graph identity, tolerance, proposal kernel, chain diagnostics, and implementation. Rhode Island supplies a reproducible low-tail example; Iowa and North Carolina supply cautionary evidence against unqualified extremum claims.

References

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- MGGG Redistricting Lab. GerryChain: A python package for redistricting ensembles. <https://github.com/mggg/GerryChain>, 2021. Version 0.3.